

Modelling & Simulation: Assignment 6

Gyroscope

Wesley Roozing, Geert Folkertsma

Introduction

In this assignment, you will learn about:

- 3D dynamics modelling,
- joints as constraints,
- rigid bodies,
- gyroscopic forces.

You will do that by modelling this:



Assignment details on second page.

Assignment

Model the gyroscope as two rigid bodies, the base as the fixed world and two joints connecting them.

You have two options for modelling the 3D dynamics of the rigid bodies:

- Extending the “house” or “Christmas tree” model for planar rigid bodies that you learned in the course Engineering System Dynamics to three dimensions.
- Using the Lie group theory that you learned in Modern Robotics or MDKS, implementing “Newton’s third law in Twists & Wrenches” in bond graphs.

For the joints, you will have to constrain the relative velocity of rotor and frame of the gyroscope; and that of the frame and the base, using:

- hard constraints (forcing equal flows);
- numerical constraints (20-sim `constrain` function); or
- physical constraints, i.e. a stiff spring-damper combination.

Some things to consider:

- What behaviour can you expect of a gyroscope?
- How do you express or represent the velocity difference that has to be constrained for the joints?
- What is the difference between the two joints?
- How do you determine (and implement) initial conditions, both for position and momentum (or velocity)?
- What is the effect of the various joint models?